

Final Project Report

Gradient Boosting Model - Home Credit Default Risk

This report summarizes the full credit default risk project. The notebook contains the step-by-step code, while this PDF shows the overall demonstration, final model result, and main evaluation evidence.

Best model hist_gradient_boosting_sklearn	Out-of-sample ROC-AUC 0.7756
Cross-validation ROC-AUC 0.7694 +/- 0.0027	Selected threshold 0.1640

Project Objective

The objective was to predict whether a loan applicant is likely to default. This is a binary classification problem where TARGET = 1 means default risk and TARGET = 0 means non-default. The main metric is ROC-AUC because the goal is to rank applicants by risk, not only classify them at one cutoff.

Data Used

The project uses the Home Credit Default Risk Kaggle data. The final run used 307,511 training applications and 315 engineered columns. The main application train/test files were used with secondary credit-history tables: bureau, previous_application, installments_payments.

Workflow

- Loaded the training and test application data.
- Checked target imbalance and missing values.
- Aggregated bureau, previous-application, and installment-payment history at the applicant level.
- Created ratio, missingness, external-score, credit-history, and payment-behavior features.
- Handled class imbalance with class-weighted models.
- Tuned model settings and compared validation AUC across model families.
- Evaluated the final model with ROC-AUC, precision, recall, F1, ROC curve, precision-recall curve, confusion matrix, and feature importance.

Model Results

Model	Validation AUC	Precision	Recall	F1
hist_gradient_boosting_sklearn	0.7756	0.2688	0.4093	0.3245

Model	Validation AUC	Precision	Recall	F1
catboost	0.7693	0.2527	0.4097	0.3126
logistic_regression_sklearn	0.7671	0.2609	0.3973	0.3150
logistic_regression_numpy	0.7641	0.2532	0.3855	0.3056
random_forest_sklearn	0.7497	0.2177	0.4598	0.2955
gradient_boosted_stumps_numpy	0.7184	0.2075	0.3771	0.2677

The gradient boosting model was selected because it produced the highest validation ROC-AUC. The final out-of-sample ROC-AUC was 0.7756, with cross-validation ROC-AUC of 0.7694 +/- 0.0027.

Feature Importance

The most important variables came from external risk scores, credit/income relationships, age and employment variables, and aggregated credit-history behavior. This is consistent with credit-risk modeling because default risk depends on both the applicant profile and past repayment behavior.

Rank	Feature	Importance
1	EXT_SOURCE_MEAN	0.07064
2	INSTAL_INSTAL_IS_LATE_MEAN	0.00599
3	BUREAU_BUREAU_DEBT_CREDIT_RATIO_RAW_MAX	0.00459
4	CREDIT_TERM	0.00438
5	AMT_ANNUITY	0.00315
6	INSTAL_AMT_PAYMENT_SUM	0.00295
7	INSTAL_AMT_INSTALMENT_MIN	0.00240
8	DAYS_BIRTH	0.00240
9	NAME_EDUCATION_TYPE=Higher education	0.00224
10	CREDIT_GOODS_RATIO	0.00190
11	DEF_30_CNT_SOCIAL_CIRCLE	0.00149
12	EXT_SOURCE_PRODUCT	0.00147

Conclusion

The project met the main objective by building an end-to-end default prediction pipeline, comparing multiple models, improving the feature set with secondary credit-history tables, and reporting a clear final ROC-AUC. The final model achieved 0.7756 out-of-sample ROC-AUC, which is substantially above random ranking and suitable for demonstrating a practical credit-risk machine learning workflow.

Repository Guide

Use [Final_Project_Report.pdf](#) for the overall project demonstration. Use [Gradient_Boosting_Model_Home_Credit_Default_Risk.ipynb](#) for the full executed notebook with code, outputs, and graphs. Supporting output tables and saved figures are in [outputs/](#) and [reports/figures/](#).